Deterministic Guardrails and Variance-Aware Evaluation for Multi-Agent Content Personalization

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Author contributions

All three authors contributed to the design, implementation and evaluation of the system described in this manuscript, and to the preparation of the manuscript itself. All authors have read and approved the submitted version and agree to be accountable for the work.

Declarations

Originality. This manuscript is original work. It has not been published previously and is not under consideration for publication elsewhere, in whole or in part.

Prior versions. No preliminary version of this work has appeared in any journal or conference. The system described here was developed as an academic project, and this manuscript is its first write-up for publication.

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Data and code availability. The source code, the 30-case evaluation dataset and every run report underlying the reported results are available from the authors on request. All figures in the manuscript are generated directly from those stored reports by a script included with the code.

Ethics. This work involved no human participants, no personal data and no animal subjects. All brand and user profiles used in the evaluation are synthetic.

Use of generative AI. Large language models are the object of study in this work and were run as part of the system and its evaluation, as described throughout the manuscript. Language models were additionally used to assist with drafting prose and reviewing source code during development. All quantitative results were produced by executing the described system and are verified against stored run artefacts; all text has been checked and is the responsibility of the authors.